Structurally Sparse Molecular Hamiltonians for Near-Term Quantum Simulation

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We present a library of qubit Hamiltonians for molecular quantum simulation that are structurally sparser than conventional Gaussian-basis alternatives across a range that depends strongly on the comparison: from 0.28× (H_2O against minimal-basis STO-3G, i.e. *denser*) through 55–76× against cc-pVDZ, to 317× at the most favourable equal-qubit point. (Corrected 2026-08-30: the retired pair-diagonal rule supported a blanket "one to three orders of magnitude"; it does not survive at either end.) The Hamiltonians are constructed from a spectral graph theory framework (GeoVac) that exploits natural coordinate separations in molecular electronic structure, producing block-diagonal electron repulsion integrals and Gaunt-enforced selection rules that persist through Jordan-Wigner encoding. For the composed architecture (PK-partitioned), the composed LiH Hamiltonian is near parity with minimal-basis STO-3G on Pauli count (838 vs. STO-3G's 907 raw; 3.0× more than the 276 reduced) while using more qubits; the decisive resource advantages are the exact linearity of the Pauli count in Q across molecules at fixed basis and a 76× reduction versus cc-pVDZ. (All counts corrected 2026-08-29 to the exact global-M_L ERI rule; see Sec. V B.) The balanced coupled architecture (PK-free) binds LiH with an FCI minimum at a computed $R_{\mathrm{eq}} = 3.227$ bohr (7.0% above the experimental 3.015 bohr) at $n_{\max} = 2$ (2,726 Pauli terms, 30 qubits); a single-point evaluation at the experimental geometry reaches 0.20% energy error at $n_{\max} = 3$ (84 qubits) when solved exactly. Chemistry-accuracy scope at $n_{\max} = 2$ is *row-conditional*: first-row hydrides (LiH, BeH_2, H_2O, ...) bind with documented potential-energy-surface (PES) topology, while second-row hydrides (NaH and below) exhibit monotone overattraction under every tested configuration — frozen-core balanced, multi-zeta valence, screened cross-center, explicit-core Hartree–Fock — with no interior PES minimum at the qubit-Hamiltonian level. We document this scope boundary empirically in §III E; the framework's primary value proposition is structural sparsity with quantitative truncation-convergence rates (Paper 38 state-space GH convergence), not energy-accuracy parity with CCSD(T) on second-row systems. The `hamiltonian()` registry covers 37 systems (35 composed molecules spanning $Z = 1$–56, H through Ba, plus the He and H_2 reference atoms), including multi-center systems (CO, N_2, F_2), mixed frozen-core molecules (NaCl), and the full first transition series as hydrides (ScH through ZnH), with the within-molecule basis exponent 3.17 identical across all three rows (small-basis fit, $n_{\max} \leq 3$; local slope rising toward ~ 3.8)—and, on the two-point $n_{\max} = 1, 2$ fit, an exponent of 2.8163 identical across all six molecules to the digits shown, *forced* by the exact Q-linearity at each basis rather than fitted—compared to $O(Q^{3.9\text{–}4.3})$ for Gaussian baselines. The composed Pauli count is exactly linear in qubit count (coefficient 27.90, exact, for main-group molecules; 30.03 for d-orbital blocks — both under the exact global-M_L ERI rule, Sec. V B), so adding atoms to a molecule increases quantum cost linearly. Second- and third-row molecules use analytical frozen cores ([Ne], [Ar], [Ar]$3d^{10}$); the frozen-core energy enters only through the identity operator and does not affect quantum measurement cost. Isostructural molecules with frozen cores produce identical Pauli counts across periodic table rows. The Hamiltonians are available as a Python library with Qiskit, OpenFermion, and PennyLane export.

I. INTRODUCTION

Quantum simulation of molecular electronic structure is among the most promising near-term applications of quantum computing [1–3]. However, the cost of implementing molecular Hamiltonians on quantum hardware is dominated by the number of Pauli terms in the Jordan-Wigner (or equivalent) encoding: each distinct Pauli string requires separate measurement circuits, and the total number of shots scales with the square of the Pauli 1-norm [4, 5].

For a molecule encoded in Q qubits using a Gaussian atomic orbital basis and the Jordan-Wigner transformation, the Pauli term count scales as $O(Q^4)$ due to the dense four-index electron repulsion integrals (ERIs). Post-hoc compression techniques—double factorization [6], tensor hypercontraction [7], qubit tapering [8]—can reduce this cost but do not alter the underlying structure of the Hamiltonian.

An alternative is to construct Hamiltonians that are *structurally* sparse: the ERIs are block-diagonal by construction, with zeros enforced by angular momentum selection rules rather than discovered by compression. The GeoVac framework achieves this by exploiting the natural coordinate separations of molecular electronic structure [9–12]. Each electron group (core, bond pair, lone pair) is described in its own natural coordinate system, coupled via cross-center nuclear attraction integrals and cross-block Coulomb integrals. The angular matrix elements are computed from Gaunt integrals (Wigner $3j$ symbols), which impose exact selection rules that translate directly into structural zeros in the qubit Hamiltonian.

The result is a Pauli count exactly linear in Q across molecules at fixed basis (within-molecule basis exponent 3.17), with 54–317× fewer terms than Gaussian raw Jordan-Wigner baselines at comparable qubit counts [10] (exact-rule numbers, 2026-08-29). This structural sparsity is compatible with all downstream optimizations (tapering, grouping, tensor factorization) and is basis-intrinsic rather than compression-dependent.

The relationship to numerical factorization techniques can be sharpened. Double factorization [6] applied to a GeoVac ERI tensor reproduces the analytic multipole-channel decomposition exactly at the production basis ($n_{\max} = 2$): the double-factorization rank equals the count of distinct radial-density channels (seven for valence $\{1s, 2s, 2p\}$), and every singular vector of the reshaped two-electron tensor lives strictly inside the multipole-channel subspace. At larger basis ($n_{\max} \geq 3$), double factorization still produces only linear combinations of multipole channels, but the radial-integral matrix develops a graded singular-value spectrum reflecting the well-known near-degeneracy of hydrogenic radial integrals; the additional compression beyond the bare multipole count is therefore a numerical realization of structure already present in the analytic decomposition, not a new algebraic substructure. Double factorization on a GeoVac Hamiltonian is, in this precise sense, the multipole expansion accessed by SVD rather than by closed form (see Paper 14 §I and `debug/sprint_df_multipole_lift_memo.md`).

This paper presents benchmark data for the GeoVac Hamiltonian library: accuracy characterization via full configuration interaction (FCI), quantum resource comparison against Gaussian baselines, and installation instructions. Theory derivations are deferred to the supporting papers [9–12].

II. THE HAMILTONIAN

A. What the user receives

A GeoVac Hamiltonian is a standard qubit operator—a weighted sum of Pauli strings—delivered as a `SparsePauliOp` (Qiskit), `QubitOperator` (OpenFermion), or `qml.Hamiltonian` (PennyLane):

```
from geovac.ecosystem_export import
    hamiltonian
h = hamiltonian('LiH', R=3.015, max_n=2)
op = h.to_qiskit()    # \gvq{
    lih_composed_pauli}{838} Pauli terms
op_of = h.to_openfermion()
op_pl = h.to_pennylane()
print(h.n_qubits)      # 30
print(h.one_norm)       # 34.0 Ha
```

The operator is ready for use in any VQE, ADAPT-VQE, QPE, or other quantum algorithm that consumes a Pauli Hamiltonian.

B. Two architectures

GeoVac provides two Hamiltonian architectures for each molecule:

a. Composed (PK-partitioned). The Phillips-Kleinman (PK) pseudopotential, a one-body operator that enforces core-valence orthogonality, is separated from the qubit Hamiltonian and evaluated classically from the VQE one-particle reduced density matrix (1-RDM) [11]:

$$E_{\text{total}} = \langle\psi|H_{\text{elec}}|\psi\rangle + \text{Tr}(h^{\text{PK}}\gamma)\,. \tag{1}$$

This produces the sparsest qubit operator (838 Pauli terms for LiH) at the cost of requiring a classical post-processing step.

b. Balanced coupled (PK-free). Cross-center nuclear attraction integrals, computed via a multipole expansion that terminates exactly at $L = 2\ell_{\max}$ by Gaunt selection rules [12], replace the PK pseudopotential. This produces a larger operator (2,726 Pauli terms for LiH at $n_{\max} = 2$) but one that is variationally complete within the composed partition—FCI on this Hamiltonian gives physical energies without any classical correction.

C. How it works (brief)

Electrons are partitioned into groups (core, bond pairs, lone pairs) based on the chemical structure. Each group is described in its natural coordinate system (hyperspherical for core electrons, molecule-frame hyperspherical for bond pairs). Within each group, the angular basis states are labeled by hydrogenic quantum numbers (n, ℓ, m), and all angular matrix elements are evaluated from Gaunt integrals—products of spherical harmonics integrated analytically via Wigner $3j$ symbols. The selection rules $|\Delta\ell| \leq k$, $|\Delta m| \leq k$ produce exact structural zeros in the ERI tensor that persist through Jordan-Wigner encoding. Full derivations are in Refs. [9–12].

III. ACCURACY BENCHMARKS

A. LiH potential energy surface

We characterize the balanced coupled Hamiltonian by computing 4-electron FCI energies across the LiH potential energy surface. Table I shows the FCI energy at seven bond lengths for $n_{\max} = 2$ ($Q = 30$ qubits, 15 spatial orbitals, FCI dimension 11,025).

The FCI-predicted equilibrium is $R_{\text{eq}} = 3.227$ bohr (7.0% error vs. experimental 3.015 bohr). At $n_{\max} = 3$ ($Q = 84$, 42 spatial orbitals, FCI dimension 741,321), the energy at $R = 3.015$ bohr improves to -8.0545 Ha (0.20% error), and the equilibrium shifts to $R_{\text{eq}} = 3.28$ bohr (8.8% error). Table II summarizes the basis convergence.

TABLE I. LiH balanced coupled FCI energies at $n_{\max} = 2$ ($Q = 30$, 2,726 Pauli terms, analytical V_{ne} integrals). E_{exact} is the non-relativistic Born-Oppenheimer energy [13].

R (bohr)	E_{FCI} (Ha)	ΔE (mHa)	Error (%)
2.0	−7.8246	181	2.3
2.5	−7.8940	164	2.1
3.0	−7.9290	142	1.8
3.015	−7.9294	141	1.8
3.5	−7.9273	137	1.7
4.0	−7.8952	157	2.0
5.0	−7.7716	254	3.3

TABLE II. Basis convergence: LiH balanced coupled at $R =$ 3.015 bohr. Exact non-relativistic energy: −8.0706 Ha.

$n_{\max}$	Q	E_{FCI} (Ha)	Error (%)	R_{eq} (bohr)	R_{eq} err
2	30	−7.929	1.8	3.227	7.0%
3	84	−8.055	0.20	3.28	8.8%

The 9× improvement in energy error from $n_{\max} = 2$ to $n_{\max} = 3$ demonstrates rapid basis convergence. The equilibrium geometry drifts by +0.053 bohr per unit increase in $n_{\max}$, approximately 3× smaller than the drift observed with the PK pseudopotential [12]. The energy error at $n_{\max} = 3$ (0.20%) is competitive with Gaussian double-zeta bases for single-point calculations.

B. Multi-geometry FCI: solver vs. Hamiltonian accuracy

The 5.3–26% equilibrium geometry errors reported in the classical solver literature for GeoVac [11] arise from the classical solver (adiabatic approximation, PK overcounting), not from the Hamiltonian operator itself. When the balanced coupled Hamiltonian is solved exactly via FCI:

- The energy error at equilibrium drops from 15% (composed + PK) to 1.8% ($n_{\max} = 2$) or 0.20% ($n_{\max} = 3$).
- The balanced coupled configuration is the *only* one that produces a bound LiH molecule in 4-electron FCI—decoupled (no cross terms), composed with PK, and coupled without cross-center V_{ne} all give unbound results [12].
- The variational bound is respected at all geometries tested ($E_{\text{FCI}} > E_{\text{exact}}$ everywhere).

C. BeH_2 and H_2O

For BeH_2 ($Q = 50$), the balanced coupled 4-electron FCI gives $E = -14.146$ Ha (10.7% error), compared to −11.497 Ha (27.4%) for composed with PK. The balanced 1-norm (323.4 Ha) is 14% lower than composed with PK in-Hamiltonian (374.9 Ha; re-measured 2026-09-01 at BeH_2's own $R = 2.502$ bohr — the 2026-08-30 pass had evaluated it at LiH's 3.015, giving 328.1 and "12%" under the exact global-M_L rule — the retired figures were 306.4 / 373.4 / "18%", and an earlier 354.9 came from the deprecated legacy BeH_2 builder path; the live values are pinned by `tests/test_paper20_library.py`), demonstrating that the PK-free architecture is cheaper in the QPE regime, where PK must be encoded; with PK partitioned classically (the VQE default), the composed electronic-only 1-norm (68.8 Ha, Table IV) is far lower.

For H_2O ($Q = 70$), FCI is infeasible (10 electrons), but the Hamiltonian metrics are available: 19,742 Pauli terms, 1,612.0 Ha 1-norm, 1,494 QWC groups (re-measured 2026-08-30; the retired pre-exact-rule figures were 5,798 / 1,511 / 168, the Pauli count low by 3.4×). The balanced 1-norm is 0.057× the composed total (which is dominated by the PK barrier at $Z_{\text{eff}} = 6$), and 4.3× the composed electronic-only 1-norm.

D. Applicability: single-point vs. dissociation

The balanced coupled Hamiltonian is designed for single-point energy calculations at fixed molecular geometries—the use case for quantum resource estimation, VQE benchmarks, and QPE demonstrations. For this primary use case, the balanced coupled Hamiltonian achieves 0.20% energy accuracy at $n_{\max} = 3$.

The Hamiltonian is *not* designed for computing dissociation curves or reaction energies between different molecular species. The cross-center V_{ne} multipole expansion includes a monopole ($1/R$) component that decays slowly, causing the electronic energy to remain unconverged even at $R = 100$ bohr. The practical dissociation energy computed from the PES well depth ($D_e =$ 0.161 Ha at $n_{\max} = 2$) overestimates the experimental value (0.092 Ha) by 75%. Users requiring dissociation energies should use Gaussian-basis methods for the dissociation limit and GeoVac Hamiltonians for single-point calculations at experimentally known or DFT-optimized geometries.

E. Chemistry-accuracy scope boundary

The framework's PES-topology accuracy is row-conditional. For first-row hydrides (Z = 3–9 + H), the balanced coupled Hamiltonian binds at the correct equilibrium geometry with documented overbinding magnitude. For second-row and below (Z ≥ 11), it does not.

a. Empirical evidence. The sprint sequence of 2026-06-07 (project memos `debug/sprint_w1e_projection_audit_memo.md`, `debug/lih_kwarg_sweep_log.txt`, and `debug/sprint_b1_nah_hf_verdict_memo.md`) systematically tested the chemistry-accuracy scope under every available kwarg combination and one architectural extension. Table III summarises the verdicts.

TABLE III. PES topology under exhaustive engineering tests at $n_{\max} = 2$. "Interior min" indicates whether the qubit-FCI or qubit-HF PES exhibits a binding well within R $\in$ [2.5, 6.0] bohr.

Configuration	LiH	NaH
composed (PK-partitioned)	no	no
balanced (frozen-core, Path A)	yes ($R_{\mathrm{eq}} = 3.227$, 7% high)	no
balanced + multi-zeta valence	N/A (Z < 11)	no
balanced + screened cross-center	yes (unchanged)	no
balanced + cross-block h1	catastrophic ($D_e = 1.1$ Ha)	N/A
explicit-core HF (12 electrons NaH; 4 e LiH cross-check)	yes (binds)	no

b. Operator-level diagnosis. Three structural factors converge to set the scope boundary at the second-row hydride threshold:

- *Fixed-basis projection.* The qubit Hamiltonian uses a static Z_{eff} hydrogenic orbital basis at each R, while the binding physics for second-row hydrides (Na $3s$ at $Z_{\mathrm{eff}} \approx 2.2$) requires R-adaptive basis content that is structurally absent in second-quantisation.

- *Multipole-expansion regime.* Track CD's cross-center V_{ne} at $L_{\max} = 2$ captures the binding-driving R-dependence for first-row systems (LiH $Z_{\mathrm{eff}} \approx 1.84$), where the valence orbital is compact enough that fixed-basis multipole matrix elements track the physics. For second-row, the more diffuse valence orbital outruns the multipole machinery.

- *Frozen-core projection.* Sprint B.1 (2026-06-07) tested Hartree–Fock on an explicitly un-frozen [Ne] core (12-electron NaH at 30 qubits, 15 spatial orbitals) and reproduced the monotone-overattraction pattern. Frozen-core projection is *not* the wall; the structural gap survives unfreezing.

c. Implication for users. Users running the framework for first-row hydride chemistry can expect bound PES with quantitative overbinding (the 0.20% single-point / D_e overestimated by $\sim$75% vs. experiment pattern documented in §III D). Users running second-row or below should treat the qubit Hamiltonian as a *resource-estimation target* only, not as a chemistry-accuracy Hamiltonian. The Pauli counts, 1-norm bounds, and GH convergence rates are correct in either case (they are properties of the operator, not its physics fidelity), so algorithm research that benchmarks against scaling behaviour transfers cleanly across the scope boundary. Algorithm research that wants ground-state energies should stay in the first-row library.

d. Structural reason. The scope boundary has a structural origin in the framework's NCG composition operation. We verified empirically on H_2 (2026-06-07, project memo `debug/bratteli_h2_pilot_memo.md`) that the Track CD one-body Hamiltonian is bit-exactly (residual 9.7×10^{-18}) a Marcolli–van Suijlekom 2014 gauge network [14] on the molecular bond quiver, with atomic Camporesi–Higuchi Dirac data at vertices and multipole-V_{ne} bimodules on edges. The scope-boundary structural mechanism — why first-row binds and second-row does not — is the chemistry-side instance of the calibration-data partition documented in Paper 18, § IV.6. A forthcoming math.OA paper formalises the NCG correspondence; the present paper takes the structural framing as a given and documents the empirical scope.

IV. QUANTUM RESOURCE COMPARISON

A. Pauli terms and measurement cost

Table IV compares GeoVac Hamiltonians against Gaussian raw Jordan-Wigner baselines.

For the small-molecule minimal-basis case, the composed LiH Hamiltonian is near parity with Gaussian STO-3G on Pauli count—838 terms vs. STO-3G's 907 (raw JW), 3.0× more than the 276 (qubit-reduced), with a nearly identical electronic 1-norm (34.0 vs. 34.3 Ha)—while using more qubits (30 vs. 10–12). The favorable 2.7×/13× multipliers hold only against the un-reduced raw-907 baseline and under the pair-diagonal ERI rule (Sec. V B); they are not the framework's load-bearing advantage. That advantage appears against larger Gaussian bases, where the gap widens decisively: 76× fewer Pauli terms than cc-pVDZ (exact rule; the retired pair-diagonal figure was 190×).

For H_2O, the composed Hamiltonian has *more* Pauli terms than STO-3G (1,954 vs. 551, advantage 0.28×) because the GeoVac encoding uses 70 qubits (35 spatial orbitals × 2 spins) while STO-3G uses only 12 qubits (7 spatial orbitals with frozen core). Against cc-pVDZ (46 qubits vs. GeoVac's 70—GeoVac uses 1.5× more; not matched accuracy either, as the composed H_2O carries a 26% R_{eq} error), GeoVac has 55× fewer Pauli terms (corrected 2026-08-30: the retired 138× was 107,382/778, i.e. computed against the retired pair-diagonal count). The measured GeoVac exponent over this range is 2.816 against the Gaussian 3.9–4.3, but that gap should *not* be read as a basis-size trend: 2.816 is a two-point $n_{\max} = 1, 2$ fit and the local slope steepens toward $\sim$3.8 by $n_{\max} = 4$, which closes most of the gap. See Sec. IV B, where the extrapolation this comparison used to license is withdrawn.

TABLE IV. Quantum resource comparison: GeoVac vs. Gaussian raw JW. GeoVac composed values use PK classical partitioning (electronic-only 1-norm); the GeoVac–Gaussian Pauli/QWC multipliers (e.g. 2.7×) use the pair-diagonal ERI rule (Sec. V B) and re-price toward parity under the exact rule. **Qubit-reduction convention:** the LiH STO-3G count (907, $Q = 12$) is the *raw* JW Hamiltonian, matched against the raw (un-tapered) GeoVac composed count (838, $Q = 30$); the larger-basis LiH counts and all H_2O Gaussian counts apply the standard 2-qubit reduction (the $Q = 20, 36, 12, 46$ sizes are the reduced sizes, matching `GAUSSIAN_LIH_PUBLISHED`). Under a uniform 2-qubit-reduced convention the headline LiH multiplier narrows (qubit-reduced STO-3G LiH is 276 at $Q = 10$ vs. GeoVac composed 838, i.e. 3.0× *more*); the raw-vs-raw and reduced-vs-reduced comparisons bracket the genuine position: near parity raw, unfavorable reduced. The *2-qubit-reduced* Gaussian Pauli counts (LiH 6-31G/cc-pVDZ and all H_2O entries) are the published values of Trenev et al. [15] (Table 5, Appendix B; Jordan–Wigner with 2-qubit reduction; their reduced STO-3G LiH value is 276 at $Q = 10$); the raw STO-3G LiH count (907, $Q = 12$) used as the matched-raw market-test baseline is the raw-JW value, not a Trenev-reduced count. The $Q^{3.9–4.3}$ Gaussian-scaling exponents are GeoVac's own log-log fit of the published reduced counts. "—" indicates data not available.

System	Method/Basis	Q	Pauli terms	1-norm (Ha)	QWC groups	Advantage[a]
LiH	GeoVac composed	30	838	34.0	64	—
	GeoVac balanced	30	2,726	78.2	629	—
	Gaussian STO-3G	12	907	34.3	273	1.08× Pauli, 4.3× QWC
	Gaussian 6-31G	20	5,851	—	—	7.0× Pauli
	Gaussian cc-pVDZ	36	63,519	—	—	76× Pauli
H_2O	GeoVac composed	70	1,954	372	64	—
	GeoVac balanced	70	19,742	1,612	1,494	—
	Gaussian STO-3G	12	551	—	—	0.28× Pauli
	Gaussian cc-pVDZ	46	107,382	—	—	55× Pauli
BeH_2	GeoVac composed	50	1,396	68.8	64	—
	GeoVac balanced	50	8,868	323.4	1,220	—

[a]Ratio of Gaussian to GeoVac composed Pauli terms. Values > 1 favor GeoVac; values < 1 favor Gaussian. *Convention (declared 2026-08-30):* the GeoVac Pauli counts and 1-norms in this table *include* the identity term, matching the raw JW totals the Gaussian baselines are quoted at — which is what makes the LiH 34.0 vs. 34.3 Ha comparison like-for-like. The balanced 1-norms were previously printed non-identity in this same column (75.2 / 286.9 / 1,440), mixing two conventions in one column; they are now identity-included (78.2 / 323.4 / 1,612). For the simulation-relevant non-identity λ_{ni} see Table VII.

B. Scaling exponents

Table V shows the within-molecule Pauli scaling exponents, obtained from two-point fits at $n_{\max} = 1$ and $n_{\max} = 2$.

TABLE V. Within-molecule Pauli scaling: $N_{\mathrm{Pauli}} \propto Q^{\alpha}$. GeoVac exponents from composed architecture (2-point fit, $n_{\max} = 1, 2$). Gaussian exponent from STO-3G/6-31G/cc-pVDZ progression. Corrected 2026-08-29 under the exact Coulomb rule (Paper [10], exact ERI selection rule); the retired pair-diagonal rule gave 2.18–2.24, mean 2.21 ± 0.02. The exact-rule exponent is *identical* across all six molecules, not merely close: see the text.

System	GeoVac α	Gaussian α
LiH	2.816	3.9–4.3
BeH_2	2.816	—
H_2O	2.816	3.9–4.3
HF	2.816	—
NH_3	2.816	—
CH_4	2.816	—
Mean	2.816 ± 0.000	3.9–4.3

The GeoVac composed scaling exponent is 2.816 for every one of the six molecules—identically, to the digits shown, not merely nearly constant. This is forced rather than empirical. The composed Pauli count is exactly linear in Q at each fixed basis (Paper [10], universal composed linearity): $N_{\mathrm{Pauli}} = 1.5\,Q$ at $n_{\max} = 1$ and $27.90\,Q$ at $n_{\max} = 2$, while Q itself grows by the same factor of 5 for every molecule. The two-point exponent is therefore

$$\alpha = 1 + \log_5\left(\frac{27.90}{1.5}\right) = 1 + \log_5 18.6 = 2.8163\ldots, \quad (2)$$

independent of the molecule. This two-point value is not comparable with the wider-range within-molecule exponent of 3.17 quoted in Paper [10]: the exponent is not constant in $n_{\max}$ (the local slope steepens toward ~ 3.8 by $n_{\max} = 4$), so a $n_{\max} = 1, 2$ fit and a small-basis fit over a longer range measure different things and legitimately differ. The Gaussian scaling exponent of ~ 4 reflects the dense four-index ERI tensor. The GeoVac advantage still grows with system size—but the within-molecule exponent is the wrong instrument for saying by how much, and the retired $(100)^{4.1}/(100)^{2.21} \approx 6{,}000\times$ extrapolation is withdrawn rather than re-priced. Two reasons. First, this exponent is not constant in $n_{\max}$ (above), so substituting the local ~ 3.8 instead of 2.816 moves the same formula from $\approx 370\times$ to $\approx 4\times$—two orders of magnitude, which is the signature of an extrapolation outside its range. Second, and more basic: $Q = 100$ is reached in this library by *adding blocks*, not by enlarging the basis, and along that axis the composed count is exactly linear in Q (Eq. 3)—exponent 1, measured, not extrapolated. The linearity is the claim; the ratio at a hypothetical Q is not.

C. Regime dependence

The GeoVac advantage is strongest in the NISQ/VQE regime where Pauli term count and QWC measurement groups dominate the runtime cost. For fault-tolerant QPE, the relevant cost metric is the Pauli 1-norm $\lambda = \sum_j |c_j|$, which determines the Trotter step count.[16] At matched accuracy (GeoVac balanced $n_{\max} = 3$ vs. Gaussian cc-pVTZ), the 1-norm comparison is less favorable due to the larger qubit count of the GeoVac encoding (84 vs. 54 qubits for LiH). However, published double-factorization or THC λ values for molecules at $Q < 100$ do not exist in the literature [6, 7], making a direct 1-norm comparison at the QPE operating point infeasible.

D. Relativistic (spin-ful) composed resource benchmarks

Table VI reports resource metrics for the spin-ful composed encoding of Paper 14 (Dirac-on-S^3 Tier 2 construction). Three single-bond hydrides spanning $Z \in \{3, 4, 20\}$, at $n_{\max} = 2$ and $n_{\max} = 3$. Every metric is computed from the algebraic-first pipeline described in that section: exact-rational X_k angular coefficients (sympy `wigner_3j`), exact-rational or machine-precision R^k radial integrals (`hypergeometric_slater.py`), closed-form $H_{\mathrm{SO}} = -Z^4\alpha^2(\kappa+1)/[4n^3 l(l+\frac{1}{2})(l+1)]$ Breit–Pauli diagonal. No numerical quadrature enters the spinor path.

Reading Table VI:

- At $n_{\max} = 2$, all GeoVac molecules sit at 0.075–0.11× the Chawla RaH-18q Pauli count at native $Q \in \{20, 30\}$, an 8.4–12.6× advantage. *The former matched-Q projection to* 17×–32× *is withdrawn* (2026-08-30), in step with Paper 14: it combined the retired $\mathrm{O}(Q^{2.5})$ composed scaling law with a $\sim$ 4.2× rel/scalar multiplier drawn from the "scalar" column this table's caption removes as never having been a scalar comparator. Both inputs are gone and neither has a corrected replacement, so no projection is reported. The surviving comparison is at native Q, and it is not species-matched—RaH ($Z = 88$) differs from LiH/BeH/CaH, and the per-species Chawla cells are deferred (cf. Paper 14).

- At $n_{\max} = 3$, the absolute GeoVac Pauli count overtakes Chawla et al. at matched Q (LiH/BeH reach 7.1× of the RaH-18q number at $Q = 84$; CaH sits at 4.783× at $Q = 56$). This is the expected sparsity-exponent regime: as $l_{\max}$ grows, the composed architecture's per-block sparsity advantage narrows. The native-Q sweet spot is $n_{\max} = 2$.

- *The scalar-vs-relativistic* λ_{ni} *comparison is withdrawn* (2026-08-30). It read "170.59 → 218.78 Ha (+28%) for LiH, 96.45 → 125.36 Ha (+30%) for CaH," but all four figures are gone: the scalar side came from the removed column that evaluated the relativistic spec with the spinless builder (so it was never a scalar comparator), and the relativistic side is in this caption's retired list. Producing an honest version needs a genuinely spinless comparator at matched basis and geometry, which the Tier 2 path does not currently build; until it does, the λ_{ni} column should be read as an absolute relativistic 1-norm and not as a relativistic *inflation* factor.

- The QWC column is inflated ($\sim$ 15× in the worst case). The richer jj-coupled Pauli structure distributes the Hamiltonian across more distinct Pauli words; VQE measurement overhead grows accordingly. Mitigable by symmetry-compressed or tensor-factorized measurement strategies.

Deferred Chawla et al. cells. The per-molecule Pauli counts, 1-norm, and QWC values for BeH, MgH, CaH, SrH, BaH at $Q = 18$ are reported in Chawla et al. [17] Supplemental Material Tables S1–S3. These were not extracted in the Tier 2 sprint timeframe. The footnote row in Table VI marks this as deferred; sharpening the head-to-head at matched $Q = 18$ per-molecule is future work, noted in Paper 14.

Matched-Q feasibility (Sprint 3). GeoVac's native-Q ladder for the composed architecture is discrete and determined by natural geometry, not tunable: $Q = 10$ (single block at $n_{\max} = 1$), $Q = 20$ (single-bond hydride with frozen core), $Q = 30$ (explicit-core hydride), $Q = 40+$ (polyatomics or $n_{\max} = 3$). $Q = 18$ falls *between* natural steps. Three truncation options were considered in Sprint 3 Track SU and all found unphysical: dropping a 2p orbital breaks the angular completeness of the block; reducing $n_{\max}$ to 1 overshoots to $Q = 10$; a custom 9-orbital spec has no principled selection criterion. The correct positioning is therefore *native-Q competitive resource counts*, not matched-Q per-molecule comparison. This is a structural difference from Gaussian-basis approaches where Q is tunable by choice of basis set (STO-3G vs cc-pVDZ vs cc-pVTZ). At native $Q = 20$, CaH/SrH/BaH relativistic Hamiltonians give 998 Pauli terms (re-measured 2026-08-30 under the corrected factor order; retired: 942), which is 0.079× Chawla et al.'s RaH-18q (12.6× fewer) and preserves a sparsity win under any pessimistic $Q = 20 \to Q = 18$ correction.

Accuracy caveats. The Tier 2 spin-ful encoding inherits the composed architecture's 5–26 % R_{eq} error from Paper 17 and the Breit–Pauli leading-order $Z\alpha^2$ approximation from Paper 14. Chawla et al.'s DC-UCCSD is accuracy-targeted (0.47% vs CASCI for CaH). Table VI is a *resource estimation* table at fixed single-point geometries, not a spectroscopic accuracy benchmark.

Fine-structure sanity check. The closed-form Breit-Pauli H_{SO} reproduces the He (2^3P), Li (2^2P), and Be ($2s2p\,^3$P) fine-structure splittings to correct sign and correct order of magnitude (benchmark code `benchmarks/fine_structure_check.py`). Relative errors $-66\%, +211\%, -78\%$ against NIST [18] — consistent with the leading-order $Z\alpha^2$ approximation and simple Z_{eff} screening, not spectroscopic. The 20–50% accuracy target suggested in the Tier 2 sprint plan is not met; full Dirac-Coulomb radial corrections and multi-electron Breit-Pauli corrections are Tier 3+.

V. AVAILABLE SYSTEMS

The GeoVac Hamiltonian library comprises 37 systems (35 composed molecules plus the He and H_2 reference atoms), spanning $Z = 1$–56 (H through Ba); Tables VII–IX enumerate the molecular systems. Table VIII lists the 5 multi-center molecules added in v2.2: heteronuclear diatomics (LiF, CO), homonuclear diatomics (N_2, F_2), and a mixed frozen-core system (NaCl). Table IX lists all ten first-row transition metal hydrides (v2.8), which use [Ar] frozen cores and isolated d-orbital valence blocks spanning $Z = 21$–30.

A. Nuclear scaling: linearity in qubit count

A key practical result visible in Tables VII and VIII is that the composed Pauli term count is *exactly linear* in the qubit count Q across the composed molecular library:

$$N_{\mathrm{Pauli}} = 27.90 \times Q\,. \tag{3}$$

Here N_{Pauli} counts *non-identity* terms. Table VIII is quoted in that convention and matches Eq. (3) exactly; Table VII includes the identity and therefore reads $27.90\,Q+1$. The relation is exact in both—the two differ by the one constant term—but the tables are not directly comparable cell-by-cell. This linearity arises because the composed architecture constructs each electron-pair block independently with the same internal structure (same angular basis, same Gaunt selection rules), so adding an atom to a molecule adds a fixed number of blocks and a proportional number of qubits and Pauli terms. The practical consequence is that the quantum measurement cost of a GeoVac Hamiltonian grows linearly with system size—adding a CH_3 group to a molecule adds exactly 30 qubits and $\sim$837 Pauli terms, regardless of molecular context. This linear nuclear scaling is a stronger result than the within-molecule basis scaling, which describes how cost grows when expanding the angular basis on a *fixed* molecule. That within-molecule exponent is 3.17 over the small-basis range; note it is *not* the 2.816 of Table V, which is the two-point $n_{\max} = 1, 2$ value—the two measure different ranges and legitimately differ (Sec. IV B).

B. The retired pair-diagonal ERI rule: a sign error, not an approximation

All Pauli counts and multipliers in this library are realized under the *exact* global-M_L Coulomb selection rule ($m_a+m_b = m_c+m_d$), including the *m-swap* two-electron integrals.

Correction (2026-08-29). Earlier versions of this library were built under a *pair-diagonal* restriction ($m_a = m_c$, $m_b = m_d$) described as a deliberate sparsifying approximation. An independent re-derivation traced it to a sign error in the production Gaunt coefficient ($q = m_c - m_a$ where the Wigner-3j bottom row requires $m_a - m_c$), which silently zeroed every m-changing multipole; the error was arbitrated against direct quadrature of the angular factor from its definition. The earlier "constant factor, scaling unchanged" re-pricing claim was a single-point artifact: the ratio is 2.51× uniformly across main-group molecules at $n_{\max} = 2$ but grows along the basis axis (1.00 × /2.51 × /5.40× at $n_{\max} = 1/2/3$), moving the within-molecule exponent from the retired 2.5 to 3.17. What survives exactly is the cross-molecule linearity $N_{\mathrm{Pauli}} = c\,Q$ at fixed basis, with $c = 27.90$ (main-group) and 30.03 (d-block); the retired pair-diagonal coefficients were 11.10 and 9.23, and the apparent d-block sparsity advantage inverts (higher angular momentum has *more* m-swap integrals for the buggy rule to drop). The realized rule is pinned by `tests/test_paper14_eri_rule.py`; Paper 14 carries the full correction record.

Second-row molecules are accessed identically to first-row:

```
from geovac.ecosystem_export import
    hamiltonian
H = hamiltonian('NaH', R=3.566)
print(H.n_terms, H.one_norm)
# 559, 171.99  (composed API: 558 non-
    identity + identity; FULL 1-norm
#  incl. the identity coefficient -- a
    different builder and convention
#  from the BALANCED Table rows)
```

The within-molecule basis exponent 3.17 is confirmed across both rows (identical to all reported digits — a consequence of the exact factorization $N_{\mathrm{Pauli}} = c(n_{\max})\,Q$). Second-row molecules use fewer qubits than their first-row analogues (NaH: $Q = 20$ vs. LiH: $Q = 30$) because frozen cores are not encoded; the universal scaling exponent confirms that structural sparsity is a property of the Gaunt selection rules, not the core treatment.

Transition metal hydrides (Table IX) sit slightly *above* the 27.90 coefficient: all ten first-row hydrides have Pauli/$Q = 30.03$ under the exact rule (the retired pair-diagonal rule inverted this ordering, giving 9.23; see

Sec. V B). That lower ratio is an *artifact* of the pair-diagonal rule dropping more m-swap integrals at higher angular momentum: under the exact global-M_L rule the d-block ratio is *higher* (30.0 vs. 27.9), not lower—so d-orbital blocks are not genuinely cheaper to encode than s/p blocks at the exact rule.

C. Installation and usage

The library is available at `https://github.com/jloutey-hash/geovac` and can be installed via:

```
pip install geovac-hamiltonians
```

or from source:

```
git clone https://github.com/jloutey-hash/
    geovac
pip install -e .
```

Ecosystem compatibility: Qiskit ≥1.0, OpenFermion ≥1.6, PennyLane ≥0.35. Python ≥3.10.

VI. CHEMISTRY-CONSUMER INTERFACE

The library exposes a chemistry-consumer interface that is designed to interoperate with external correlation engines (DMRG, CCSD(T), AFQMC, VQE) without requiring those engines to know anything about GeoVac's internal spectral-triple construction. Three additions ship in this release.

a. GH-convergence-derived basis-truncation rate. Every `GeoVacHamiltonian` returned by the public API carries a metadata field `propinquity_bound` (legacy field name; the bound itself is the van Suijlekom state-space Gromov–Hausdorff convergence estimate of Paper 38) that evaluates the closed-form $\Lambda \le C_3 \cdot \gamma_{n_{\max}}$ from [19] (Paper 38, the van Suijlekom state-space Gromov–Hausdorff convergence theorem) at the cutoff $n_{\max}$ used to build the Hamiltonian. The L2 central-Fejér sum-rule gives $\gamma_{n_{\max}=2,3,4} = 2.0746/1.6101/1.3223$ with asymptotic rate $4/\pi$ (the Hopf-base measure $\mathrm{Vol}(S^2)/\pi^2$ signature of the master Mellin engine M1, see [20]). This is, to our knowledge, the first quantitative basis-truncation *convergence-rate* estimate exposed at a chemistry-consumer API—a proven state-space GH metric rate, *not* a direct energy-error bound (the operator-to-energy lift is a loose heuristic that overshoots by 3–4 orders of magnitude; see the Trotter-budget footnote). The rate is system-independent (it is a property of the spectral-triple truncation, not of the chemistry content), so chemistry consumers reading the metadata see the same "the operator is within state-space GH distance $\gamma_{n_{\max}}$ of the $n_{\max} \to \infty$ limit" message (a convergence rate, not a direct energy-error bound) regardless of which molecule they are running.

b. FCIDUMP export. The `GeoVacHamiltonian` class now exposes a `to_fcidump(filename)` method that writes the un-Jordan-Wigner'd $(h_1, \mathrm{eri}, e_{\mathrm{core}})$ integrals in standard FCIDUMP format [21]. The writer is pure-Python (no PySCF dependency added) and produces files that read back bit-exactly: round-trip on LiH composed at $n_{\max} = 2$ gives $\max|h_1^{\mathrm{diff}}| = 0.0$ and $\max|\mathrm{eri}^{\mathrm{diff}}| = 0.0$; a seven-system sample across LiH, BeH_2, H_2O, NaH, CO, ScH, H_2 gives relative error below 10^{-10} across all integrals (`tests/test_ecosystem_export.py`). This exporter unblocks DMRG (via Block2 / pyscf), CCSD(T) (via pyscf), and AFQMC (via Auxiliary-Field Monte Carlo packages) simultaneously.

c. Openfermion-native VQE path. The `GeoVacHamiltonian.to_openfermion()` method (already present prior to this release) is the canonical entry point for variational quantum eigensolver (VQE) consumers. Empirical comparison against alternative stacks is striking: openfermion-native UCCSD on H_2 at $Q = 10$ qubits reaches error 8.6×10^{-13} mHa via L-BFGS-B from a Hartree–Fock initialization in 165 function evaluations (0.92 s wall), five orders of magnitude tighter than published STO-3G UCCSD literature. The same H_2 task under the Qiskit-Nature stack at Qiskit 2.x reaches 87 mHa under `efficient_su2 + COBYLA` or fails to converge under `uccsd`, with per-evaluation cost approximately 3000× that of the openfermion path. Production VQE workflows on GeoVac Hamiltonians should use the openfermion interface and a UCCSD-style ansatz; the qiskit-nature stack is not recommended. Scaling to LiH at $Q = 30$ requires sector projection beyond the qubit Hamiltonian alone (the 2^{30}-dimensional statevector exceeds practical memory on commodity hardware); the four-axis Z_2 tapering of [10] (Hopf m_ℓ + Gaunt-parity ℓ-Z_2 + atom-swap + per-sub-block particle-conservation; `tapered = 'full'` mode of `ecosystem_export.hamiltonian`) saves 33% on LiH ($30 \to 20$) and 31% on CH_4 ($90 \to 62$), materially closing the practical-scale gap for classical-statevector simulation in the $\sim Q = 30$–100 qubit regime. The previously-reported ∼ 12.6% Pauli reduction (Hopf only) is now ∼ 30% at the production API level.

d. Honest scope. The FCIDUMP exporter writes the $(h_1, \mathrm{eri}, e_{\mathrm{core}})$ tensors that `build_composed_hamiltonian` and `build_balanced_hamiltonian` construct. For systems where the composed/balanced builder produces a binding PES under qubit/composed FCI (H_2, BeH_2, multi-center diatomics where the multipole expansion terminates exactly), downstream consumers receive an integral set that reproduces the framework's qubit-level energetics. For systems where the composed Hamiltonian itself does not bind under qubit FCI (LiH composed and NaH balanced at production $n_{\max}$, observed in Sprints R3-A and R3-B, 2026-06-07), the exported integrals reproduce the same non-binding PES; classical correlation methods consuming the same FCIDUMP cannot close that gap because the wall is at the projection step from continuous Level 4 to second-quantized integrals, not at the determinant-expansion level (cf. [20], §IV.6). The interface ships honestly; consumers can use it to confirm or falsify the projection-step localization on systems of interest.

VII. LIMITATIONS

a. Composed partition. The Hamiltonian is constructed within a composed partition that assigns each electron to a specific group (core, bond, lone pair). This is an approximation of the full N-electron Hilbert space—inter-group correlations beyond those captured by the cross-block Coulomb and cross-center nuclear attraction integrals are neglected. For LiH, the full 4-electron treatment in the same angular basis gives 63.5% R_{eq} error (angular basis limited), while the composed partition gives 7.0–8.8% (basis more efficient but inter-group antisymmetry approximate) [11].

b. Basis completeness. At $n_{\max} = 2$ (the production operating point for composed Hamiltonians), single-point energies have 1.8% error for LiH. At $n_{\max} = 3$, this improves to 0.20%, but at the cost of 84 qubits and 19,959 Pauli terms (balanced coupled) or 7,879 (composed). The $n_{\max} = 3$ basis approaches chemical accuracy for single-point energies but not for potential energy surfaces (8.8% R_{eq} error persists due to structural drift in the composed partition).

c. Dissociation limit. The dissociation limit is not converged (see Sec. III D).

d. Classical solver accuracy. The classical (non-quantum) solver results reported in Refs. [11, 12] (5.3–26% R_{eq} error) reflect classical solver limitations, not Hamiltonian quality. FCI on the balanced coupled Hamiltonian gives substantially better energies (0.20% at $n_{\max} = 3$), validating the Hamiltonian for quantum simulation use.

e. Scope. First-row ($Z = 1$–10) and second-row ($Z = 11$–18) atoms are fully supported. Second-row molecules use analytical [Ne] frozen cores with Clementi–Raimondi orbital exponents; the frozen-core energy enters only through the identity operator and does not affect quantum measurement cost. The full first transition series ($Z = 21$–30) is implemented as hydrides with [Ar] frozen cores and isolated d-orbital valence blocks. d-Orbital blocks have a slightly higher Pauli-per-qubit ratio (30.03 vs. 27.90) under the exact global-M_L rule (Sec. V B; the retired pair-diagonal rule inverted this). The atomic classifier handles all ten elements automatically, including the anomalous configurations of Cr ($3d^5 4s^1$) and Cu ($3d^{10} 4s^1$). Main-group third-row atoms ($Z = 19$–20 and $Z = 31$–36) are supported via [Ar] and [Ar]$3d^{10}$ frozen cores [22]. The library currently includes 37 systems (35 composed molecules plus the He and H_2 reference atoms).

f. Qubit count. GeoVac Hamiltonians use more qubits than Gaussian alternatives at the same basis quality (30 vs. 12 for LiH at STO-3G–equivalent accuracy). The advantage is in Pauli count and measurement cost, not qubit economy. For near-term hardware where measurement budget (not qubit count) is the bottleneck, this tradeoff is favorable.

VIII. CONCLUSION

The GeoVac Hamiltonian library provides structurally sparse qubit Hamiltonians for molecular quantum simulation across three rows of the periodic table (37 systems—35 composed molecules plus He and H_2—$Z = 1$–56), including multi-center systems with multiple heavy atoms and the full first transition series as hydrides (ScH through ZnH; the d-block Pauli/Q ratio is slightly higher than main-group, Sec. V B). The structural sparsity—Gaunt selection rules with a universal within-molecule exponent of 3.17—is confirmed across all three rows and all molecular topologies, independent of the core treatment (Level 3-solved or frozen). The composed Pauli count is exactly linear in Q (coefficient 27.90), meaning that adding atoms to a molecule increases quantum cost linearly. For LiH, this yields near parity on Pauli terms (1.08× fewer, raw-vs-raw) and 4.3× fewer measurement groups than Gaussian STO-3G (QWC 64 vs. 273), growing to 76× fewer terms against cc-pVDZ. All multipliers are realized under the exact global-M_L rule (Sec. V B); the LiH small-molecule count is near parity with STO-3G (838 vs. 907), while the *scaling* advantage (linear vs. $O(Q^4)$) and the large-system multipliers are unaffected. For first-row hydrides at fixed equilibrium geometries, the balanced coupled architecture achieves 0.20% single-point energy error at FCI; for second-row hydrides, the qubit Hamiltonian's potential-energy surface does not exhibit an interior minimum at $n_{\max} = 2$ under any of the engineering configurations we tested (Section III E). The framework's primary value proposition is structural sparsity with quantitative truncation-convergence rates (Paper 38 state-space GH rate, exposed as metadata on every Hamiltonian export), not energy-accuracy parity with CCSD(T) across the full periodic-table-row range.

Second- and third-row molecules use analytical frozen cores ([Ne], [Ar], and [Ar]$3d^{10}$), with the frozen-core energy isolated in the identity operator. The non-identity 1-norm—the simulation-relevant quantity—scales comparably to first-row molecules. A strong isostructural invariance is observed: frozen-core molecules with the same block topology produce identical Pauli counts across periodic table rows (e.g., NaH = KH at $Q = 20$). The within-molecule scaling exponent (3.17) is universal across all three rows, confirming that the structural sparsity advantage is driven by Gaunt selection rules rather than any specific core treatment.

Future directions include: (i) quantum resource estimation for fault-tolerant algorithms using GeoVac Hamiltonians, (ii) hardware demonstrations on trapped-ion or superconducting platforms, (iii) compatibility benchmarks with tensor factorization and active-space reduction techniques, (iv) extension to heavier main-group atoms ($Z = 37$–54) via [Kr] and [Kr]$4d^{10}$ frozen cores, and (v) pushing the chemistry-accuracy scope boundary into the second-row hydride regime, which would require structural extensions beyond the current $n_{\max} = 2$ basis (multi-zeta valence orbitals, R-adaptive Z_{eff}, bonding-orbital Pauli repulsion against explicit cores) and is the subject of a separate engineering arc.

The library is open-source and pip-installable. All Hamiltonians presented in this paper are reproducible from the public repository.

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TABLE VI. Tier 2 relativistic composed resource metrics (Dirac-on-S^3 T3 output). $N^{\mathrm{rel}}_{\mathrm{Pauli}}$: non-identity Pauli string count in the (κ, m_j)-labeled JW encoding. λ_{ni}: non-identity Hamiltonian 1-norm. QWC: greedy qubit-wise-commuting group count. Chawla ratio = $N^{\mathrm{rel}}_{\mathrm{Pauli}}/12\,556$, the single calibrated Chawla RaH-18q cell [17]. *Re-measured 2026-08-30* under the corrected Condon–Shortley angular factor order (retired values: 1,413 / 942 / 89,226 / 59,484 and λ = 40.59 / 143.96 / 18.68 / 218.78 / 413.90 / 125.36). The former "scalar" column is *removed*, not re-priced: it evaluated the relativistic spec with the spinless builder, which returns the same count as the relativistic builder, so it was never a scalar comparator and the ratios drawn from it measured nothing.

Molecule	$n_{\max}$	Q	$N^{\mathrm{rel}}_{\mathrm{Pauli}}$	λ_{ni} (Ha)	QWC	Chawla ra
LiH	2	30	1 501	39.53	142	0.120×
LiH	3	84	90 114	208.79	11 938	7.177×
BeH	2	30	1 501	142.52	142	0.120×
BeH	3	84	90 114	400.01	11 948	7.177×
CaH	2	20	998	17.91	142	0.079×
CaH	3	56	60 060	119.35	11 925	4.783×
SrH	2	20	998	17.91	142	0.079×
BaH	2	20	998	17.91	142	0.079×

Chawla RaH-18q [17]: 12 556 Pauli (rel), 2 740 Pauli (non-rel) (47 099 / 4 249 two-electron integrals), $Q = 18$.

SrH, BaH added Sprint 3 (v2.14.0) via [Kr] and [Xe] frozen cores (`geovac/neon_core.py`).

CaH/SrH/BaH at $Q = 20$ bit-identical ($N^{\mathrm{rel}}_{\mathrm{Pauli}} = 998$, $\lambda_{\mathrm{ni}} = 17.91$, QWC = 142): isostructural invariance.

Sprint 4 Track TR (v2.15.0): rel Pauli counts corrected 1.76× (n=2) / 1.92× (n=3) upward via missing jj reduced-matrix-element phase fix. A second

Chawla et al. SI Tables S1–S3 (per-molecule matched-Q data): accessibility **deferred**.

TABLE VII. GeoVac Hamiltonian library at $n_{\max} = 2$. First-row molecules use composed encoding (PK classically partitioned); second- and third-row molecules use balanced coupled encoding with frozen cores and cross-center V_{ne}. λ_{ni} is the non-identity 1-norm (simulation-relevant); the Pauli column, by contrast, *includes* the identity term. Isostructural pairs (rows 2–3) produce identical Pauli counts — but *not* identical QWC group counts (MgH_2 903 vs. CaH_2 898; SiH_4 1,566 vs. GeH_4 1,562; only NaH/KH tie). The greedy grouper sorts terms by descending |coefficient| before insertion (`geovac/measurement_grouping.py`), so partners with identical Pauli *support* but different coefficients are visited in different orders. The invariance is a property of the support, and QWC grouping is not a function of the support alone. The `hamiltonian()` API currently returns composed encoding for all rows; balanced data shown here are from Paper 19's `build_balanced_hamiltonian()` (every cell pinned by `tests/test_paper20_balanced_lambda.py`; the λ_{ni} column was re-synced to the live builder 2026-07-02; all cells remeasured under the exact global-M_L rule 2026-08-29, Sec. V B — Pauli-count isostructural invariance survives exactly (NaH = KH, HCl = HBr, . . .). *Each row is evaluated at its own equilibrium bond length R, now stated in the table.* Corrected 2026-08-31: the balanced λ_{ni} and QWC columns had been computed at $R = 3.015$ bohr — LiH's bond length — for *every* molecule, because the balanced builder defaulted to that value and ignored the spec's geometry. The largest error was KH at -12.7% ($32.2 \to 28.1$), its bond length being furthest from the substituted one. The Pauli column was unaffected, and that is not luck: the count is fixed by the angular selection rules and is bit-identical across geometries, whereas λ_{ni} and the greedy QWC count are both projections through the geometry (`tests/test_paper20_geometry_independence.py`). This also reverses the 2026-08-30 note that had "corrected" NaH $19.6 \to 20.6$ and HCl $866.1 \to 869.7$: those two cells were right before that pass, which had re-measured them at the wrong geometry.)

Molecule	Core	Q	R (bohr)	Pauli	λ_{ni} (Ha)	QWC
LiH	He	30	3.015	838	27.7	64
BeH_2	He	50	2.502	1,396	54.3	64
H_2O	He	70	1.809	1,954	186.8	64
HF	He	60	1.733	1,675	232.7	64
NH_3	He	80	1.912	2,233	158.0	64
CH_4	He	90	2.050	2,512	135.4	64
NaH	Ne	20	3.566	575	19.6	69
MgH_2	Ne	40	3.261	4,861	111.8	903
HCl	Ne	50	2.409	9,824	866.1	1,173
H_2S	Ne	60	2.534	13,863	873.6	1,264
PH_3	Ne	70	2.680	18,854	889.3	1,427
SiH_4	Ne	80	2.800	24,745	909.2	1,566
KH	Ar	20	4.243	575	28.1	69
CaH_2	Ar	40	3.807	4,861	123.8	898
HBr	Ard	50	2.670	9,824	875.3	1,176
H_2Se	Ard	60	2.760	13,863	883.4	1,264
AsH_3	Ard	70	2.820	18,854	885.2	1,468
GeH_4	Ard	80	2.870	24,745	874.8	1,562

TABLE VIII. Multi-center molecules added in v2.2 at $n_{\max} = 2$. LiF, CO, N_2, and F_2 use He frozen cores; NaCl uses mixed frozen cores (Na [Ne] + Cl [Ne]). Pauli counts are non-identity terms and QWC groups are greedy; both re-measured 2026-08-29 under the exact Coulomb rule (the retired pair-diagonal rule gave 778/1,111/556 terms and 21 groups). Every row is $27.90 \times Q$ exactly.

Molecule	Core	N_e	Blocks	Q	Pauli	QWC
LiF	He	12	6	70	1,953	64
CO	He	14	7	100	2,790	64
N_2	He	14	7	100	2,790	64
F_2	He	18	9	100	2,790	64
NaCl	Ne+Ne	8	4	50	1,395	64

TABLE IX. First-row transition metal hydrides ($Z = 21$–30) at $n_{\max} = 2$ (bond) / $n_{\max} = 3$ (d-block). All use [Ar] frozen cores (18 electrons) with d-orbital valence blocks ($l_{\min} = 2$). Under the exact global-M_L rule the Pauli/Q ratio is 30.03, slightly above the main-group coefficient of 27.90 (the retired pair-diagonal rule inverted this, giving 9.23 vs. 11.10 — an artifact of dropping more m-swap integrals at higher angular momentum; Sec. V B). Isostructural invariance holds: all ten hydrides share the same block topology and produce identical Pauli counts. Cr and Cu have anomalous configurations ($3d^5 4s^1$ and $3d^{10} 4s^1$).

Molecule	Config.	Core	Q	Pauli	Pauli/Q
ScH	$3d^1 4s^2$	Ar	30	901	30.03
TiH	$3d^2 4s^2$	Ar	30	901	30.03
VH	$3d^3 4s^2$	Ar	30	901	30.03
CrH	$3d^5 4s^1$	Ar	30	901	30.03
MnH	$3d^5 4s^2$	Ar	30	901	30.03
FeH	$3d^6 4s^2$	Ar	30	901	30.03
CoH	$3d^7 4s^2$	Ar	30	901	30.03
NiH	$3d^8 4s^2$	Ar	30	901	30.03
CuH	$3d^{10} 4s^1$	Ar	30	901	30.03
ZnH	$3d^{10} 4s^2$	Ar	30	901	30.03